Beyond the Scale: Modeling Anastomotic Leak as BMI Increases

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01

Model Selection

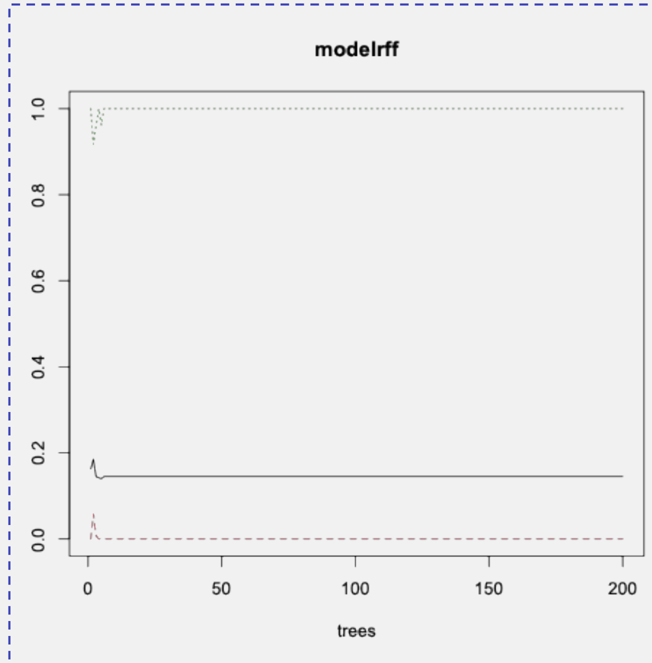
Using the Random Forest is one of the best for this model

Creating the Model

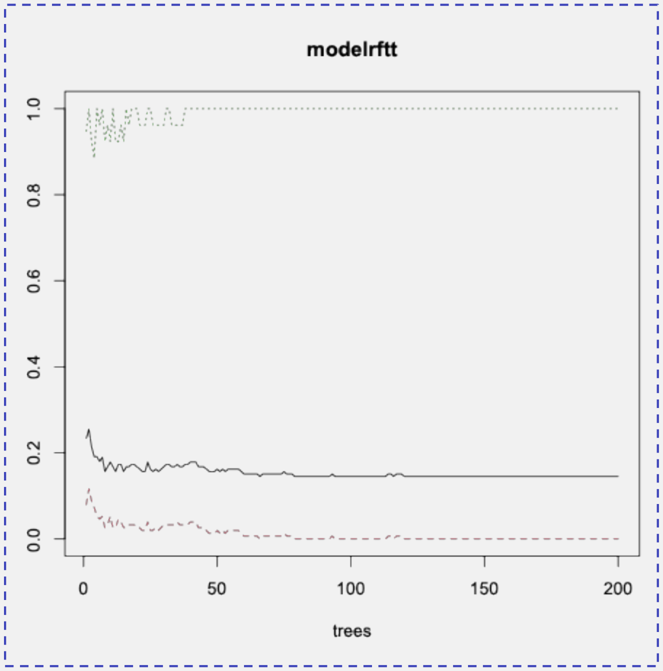
- Data set includes information on 179 patients
- Predictors include: age, gender, race, height, weight, tobacco usage, diabetes, probability, coronary or pulmonary artery disease predictors, cancer probability, albumin measure, operation length in days, and BMI levels
- All categorical predictors and response variables were converted to factors
- can build a Random Forest classification model using patient demographic and clinical predictors
- Keep response variable for anastomotic leaking as a binary outcome (0 = no leak, 1 = leak)

Why Random Forest

- method that improves accuracy
- reduces overfitting relative to a single decision tree
- tree-based methods are suited for these complexities without requiring explicit specification of interactions or transformations
- has high variance and can overfit small medical datasets



Keep it in Tune!



- set the number of trees (ntree) to 200, used mtry = 2 to control the number of randomly selected predictors at each split, and increased nodesize to prevent overly small terminal nodes
- tuned random forest model outperforms the original model
- tuned model predicts leak probability more accurately and with less variability

02

Effects of BMI

How BMI effects the probability of Anastomotic Leaking

BMI -

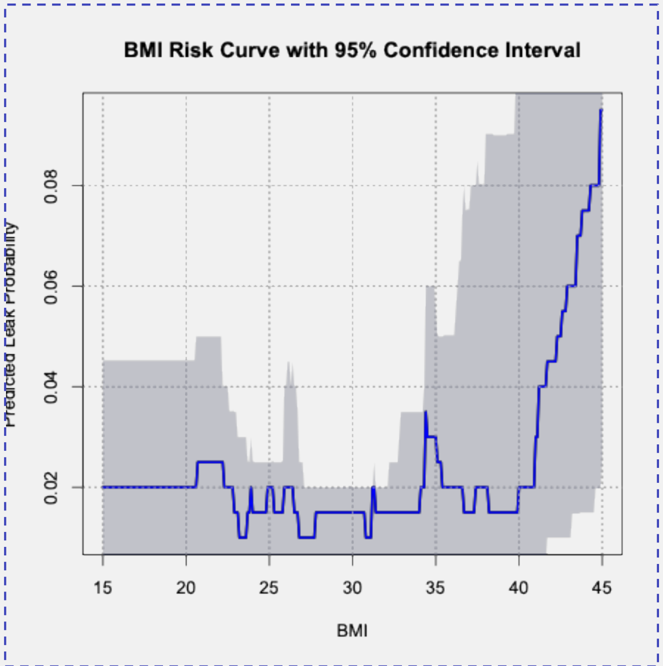
- BMI is a key predictor in determining the possible risk of anastomotic leak after a colectomy operation
- approach isolates the impact of obesity while preserving each patient’s unique clinical profile
- Resulting curves offers insight into how BMI interacts with age, comorbidities, albumin levels, and operative time to influence the probability of an anastomotic leak
- BMI level chart:
    - < 18.5 = underweight
    - 18.5 – 24 = normal
    - 25 – 35 = overweight
    - 35 ≤ = obese

02.1 Arizona Robbins

35-year-old female with no comorbidities, a post-operative albumin level of 4.2 and an operation length of 90 minutes (0.0625 fraction of a day)

Lower Comorbidities, Lower Probability

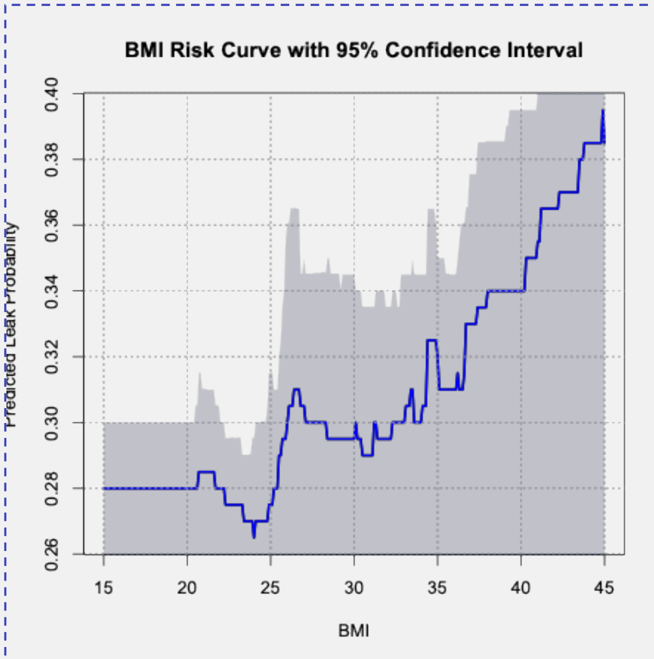
- Levels 15-21: remains constant of 2%
- Levels 22-33: can range as high as less than 4% or as low as 0-1%
- Levels 34+: increasing to a high of about 10% probability
- represents a very low-risk patient resulting in a better, near perfect, post operation outcome
- BMI does not meaningfully increase risk until BMI exceeds approximately 35



02.2 Richard Webber

62-year-old African American male with tobacco use, diabetes, low albumin levels of 2.8, and a long operative time of 210 minutes (0.1458 fraction of a day)

Higher Comorbidities, Higher Probability



- Levels < 21: probability of an anstomotic leaking post operation stays at 28%
- Levels 22-25: decreases gradually about two percent to at lowest 26%
- Levels 25-27: rises back up to almost 30% probability
- Levels 30+: probability gradually increases and goes as high as almost 40%
- curve demonstrates that obesity sharply increases the danger of anastomotic leak for patients with multiple comorbidities

Thank You

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